

Using ChatGPT, find a single command and its command-line flags that, when executed, lists all **TCP** sockets in a **LISTEN** state on an **IPv4** address, showing the **program** that is using it. Note that you will need to run this command using **sudo** since administrator privileges are required to list resources being used by other programs.

- Take a screenshot of the prompt and the command that ChatGPT generates

You can use the **ss** command to list all TCP sockets in a LISTEN state on an IPv4 address, showing the program using it. Here's the command:

```
bash
```

[Copy code](#)

```
sudo ss -tlnp
```

Explanation:

- **ss** : The utility to display network socket information.
- **-t** : Restricts the output to TCP sockets.
- **-l** : Lists sockets in the LISTEN state.
- **-n** : Displays numeric addresses and ports (without DNS name resolution).
- **-p** : Shows the process using the socket (requires **sudo** for processes owned by other users).

This command will provide a concise view of all listening TCP sockets, their IPv4 addresses, and the associated programs.

- Run the command using **sudo** and take a screenshot of the output to include in your lab notebook.

```
jgafron@course-vm:~$ sudo ss -tlnp
State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
LISTEN  0        128     0.0.0.0:22        0.0.0.0:*          users: ( ("sshd",pid=819,fd=3) )
LISTEN  0        4096    127.0.0.53:53     0.0.0.0:*          users: ( ("systemd-resolve",pid=398,fd=14) )
LISTEN  0        4096    127.0.0.1:44879   0.0.0.0:*          users: ( ("containerd",pid=479,fd=9) )
LISTEN  0         2       [::]:3350        [::]:*            users: ( ("xrdp-sesman",pid=519,fd=7) )
LISTEN  0        128     [::]:22          [::]:*            users: ( ("sshd",pid=819,fd=4) )
LISTEN  0         2       *:3389           *:*               users: ( ("xrdp",pid=662,fd=11) )
```

- List a service that can be contacted from any interface on the machine. List a service that can only be contacted by local processes.

A service that can be contacted from any service on the machine would be "xrdp" hence the star under the local address.

A service that can only be contacted by local processes would be “systemd-resolve” and “containerd”. You can tell this by the loopback address listed as the services local address.

- Run the command again, but do not use `sudo` as this is a machine managed by CAT. Include a screenshot of the output.

```
jgafron@ada:~$ ss -tlnp
State      Recv-Q    Send-Q    Local Address:Port    Peer Address:Port    Process
LISTEN     0          100       127.0.0.1:25          0.0.0.0:*             xinetd
LISTEN     0          4096      127.0.0.1:631        0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6102       0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6103       0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6100       0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6101       0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6108       0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6106       0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6107       0.0.0.0:*             xinetd
LISTEN     0          128       127.0.0.1:6105       0.0.0.0:*             xinetd
LISTEN     0          4096      127.0.0.53%lo:53     0.0.0.0:*             systemd-resolve
LISTEN     0          1024      127.0.0.1:42179      0.0.0.0:*             containerd
LISTEN     0          4096      127.0.0.54:53        0.0.0.0:*             xinetd
```

- Take a screenshot of the prompt and the command that ChatGPT generates

```
bash

sudo lsof -iTCP -sTCP:LISTEN -n -P
```

 Copy code

- Run the command using sudo and take a screenshot of the output to include in your lab notebook.

```
jgafron@course-vm:~$ sudo lsof -iTCP -sTCP:LISTEN -n -P
COMMAND  PID  USER      FD  TYPE DEVICE SIZE/OFF NODE NAME
systemd-r 398 systemd-resolve 14u IPv4  4143    0t0  TCP 127.0.0.53:53 (LISTEN)
container 479  root      9u  IPv4  5052    0t0  TCP 127.0.0.1:44879 (LISTEN)
xrdp-gesm 519  root      7u  IPv6  5326    0t0  TCP [::1]:3350 (LISTEN)
xrdp      662  xrdp     11u IPv6  5520    0t0  TCP *:3389 (LISTEN)
sshd      819  root      3u  IPv4  5764    0t0  TCP *:22 (LISTEN)
sshd      819  root      4u  IPv6  5766    0t0  TCP *:22 (LISTEN)
```

- Show a screenshot of the measured bandwidth available between your us-west1-b VM and each of the other Compute Engine VMs. Explain the relative differences (or lack thereof) in your results.

australia-southeast

```
jgafron@vm-us-west1-b:~$ sudo iperf -c 10.152.0.2 -p 80
-----
Client connecting to 10.152.0.2, TCP port 80
TCP window size: 85.0 KByte (default)
-----
[  1] local 10.138.0.6 port 43488 connected with 10.152.0.2 port 80
[ ID] Interval           Transfer     Bandwidth
[  1] 0.0000-10.1552 sec   190 MBytes  157 Mbits/sec
```

europa-west

```
jgafron@vm-us-west1-b:~$ sudo iperf -c 10.132.0.2 -p 80
-----
Client connecting to 10.132.0.2, TCP port 80
TCP window size: 85.0 KByte (default)
-----
[  1] local 10.138.0.6 port 52296 connected with 10.132.0.2 port 80
[ ID] Interval           Transfer     Bandwidth
[  1] 0.0000-10.1972 sec   175 MBytes  144 Mbits/sec
```

us-east

```
jgafron@vm-us-west1-b:~$ sudo iperf -c 10.142.0.2 -p 80
-----
Client connecting to 10.142.0.2, TCP port 80
TCP window size: 85.0 KByte (default)
-----
[  1] local 10.138.0.6 port 43666 connected with 10.142.0.2 port 80
[ ID] Interval           Transfer     Bandwidth
[  1] 0.0000-10.1284 sec   393 MBytes  325 Mbits/sec
```

Name	Status	Type	Initiator	Size	Time
www.google.com	200	docum...	Other	61.7 kB	117 ms
m=cdo\$,\$sm,\$sa,mb4ZUb,cEt9...	200	stylesh...	(index):16	1.7 kB	18 ms
m=cdo\$,\$sm,\$sa,mb4ZUb,cEt9...	200	script	(index):16	337 kB	62 ms
googlelogo_color_272x92dp.p...	200	png	(index):63	6.0 kB	32 ms
hpba?yv=3&cs=0&ei=kAOGZ...	200	xhr	(index):69	152 B	65 ms
data:image/png;base...	200	png	(index):117	(memor...	0 ms
rs=AA2YrTvH0Rknr6hXqx-tgqA...	200	script	(index):114	81.4 kB	187 ms
rs=AA2YrTvDtorsWuiBHYzP5-l...	200	stylesh...	(index):114	2.8 kB	150 ms
desktop_searchbox_sprites318...	200	webp	(index):120	682 B	31 ms
gen_204?s=webhp&t=aft&aty...	204	ping	(index):14	17 B	27 ms
gen_204?atyp=csi&ei=kAOGZ...	204	ping	m=cdo\$,\$sm,\$sa,m	17 B	27 ms
gen_204?s=async&astyp=hp...	204	text/html	(index):3	17 B	24 ms
search?q&cp=0&client=gws-w...	200	xhr	m=cdo\$,\$sm,\$sa,m	410 B	48 ms
m=sb_wiz,aa,abd,U9EYge,sy19...	200	script	m=cdo\$,\$sm,\$sa,m	118 kB	24 ms
m=Wo3n8,syro,loL8vb,sysc,sys...	200	script	m=cdo\$,\$sm,\$sa,m	42.0 kB	20 ms
rs=ACT90oERHbyrwqvMgvT_1...	200	fetch	m=cdo\$,\$sm,\$sa,m	1.0 kB	20 ms
client_204?atyp=i&biw=1365...	204	text/html	(index):3	21 B	29 ms
m=syjk,synp?xjs=s4	200	fetch	m=cdo\$,\$sm,\$sa,m	805 B	23 ms
m=sy1cu,P10Owf,sy1bm,sy1bk...	200	script	m=cdo\$,\$sm,\$sa,m	8.2 kB	22 ms
gen_204?atyp=csi&ei=kAOGZ...	204	ping	m=cdo\$,\$sm,\$sa,m	21 B	31 ms

- What is the URL being requested?
- Explain the HTTP status code that is returned and what the code indicates
- Take a screenshot indicating the version of the HTTP protocol that is used for each request.
(Hint: look at the response status line and **alt-svc:** HTTP response headers indicating HTTP/2 or HTTP/3).

URL requested is google.com

It returned 200 OK, which is the success code. It means the request was successful and the URL was fetched.

HTTP/2

No location header is present

- Take a screenshot of when cookies are set via **Set-Cookie:**

- Take a screenshot of when cookies are attached via [Cookie:](#)

```
Set-Cookie: AEC=AZ6Zc-
            WPrzTZokDvxKDtSpdBKLZIRGsK9MpOB51vuHlOnjW3ugGbiUy
            MvWA; expires=Sun, 13-Jul-2025 06:26:24 GMT; path=/;
            domain=.google.com; Secure; HttpOnly; SameSite=lax

Set-Cookie: NID=520=DhRsEuBgIHQ8ctBphXA-
            F6Vd7ASkKS4JSKKv8Gb6UgSuGZov9NwDbv1rP4flb5IZ6kx35uG
            N7PcL-
            QA1WEoy9DeMlwO7rZfguktx3tPpbZJn2jsfWx33j62pNKQ7Aqw
            m1h_FdOvmh7wzqtPxqKZhyUH8Tb7th468VpeqlZVIF3VUjnurkl
            cxPRV3qY6vVRzDSjUfNIUm_7oXgk6FQ; expires=Wed, 16-Jul-
            2025 06:26:24 GMT; path=/; domain=.google.com; Secure;

Cookie: AEC=AZ6Zc-
        WPrzTZokDvxKDtSpdBKLZIRGsK9MpOB51vuHlOnjW3ugGbiUyMvW
        A; NID=520=DhRsEuBgIHQ8ctBphXA-
        F6Vd7ASkKS4JSKKv8Gb6UgSuGZov9NwDbv1rP4flb5IZ6kx35uGN7Pc
        L-
        QA1WEoy9DeMlwO7rZfguktx3tPpbZJn2jsfWx33j62pNKQ7Aqwm1h
        _FdOvmh7wzqtPxqKZhyUH8Tb7th468VpeqlZVIF3VUjnurklcxPRV3q
        Y6vVRzDSjUfNIUm_7oXgk6FQ
```

- Show the requests and responses in the listing. Click on the last request sent, then click on the response to see that its payload has returned the data that is then rendered on the search page similar to what is shown below for ["rabbid"](#)

Name	X	Headers	Payload	Preview	Response	Initiator	Timing	Cookies
rs=ACT90oEcwjFbF4BR23XfuVr...	1	)						
GetAsyncData	-	]						
search?q&cp=0&client=gws-...	-	}]						
search?q=Portland%20State&...	2	[[["\u003cb\u003ePortland state university acceptance rate\u003c\b\u003e",						
m=sy1ir,syqv,syqu,syqw,sy1j6,s...								
asyncContextualTask?vet=12a...								
bgasy?ei=HQWGGZ7KMBpLB0P...								
vpkg?vet=10ahUKEwjyp3cy_S...								
log?format=json&hasfast=true								
asyncContextualTask?vet=12a...								
asyncContextualTask?vet=12a...								